Exercise: Explaining the increase

Introduction

One of the essential tasks of the marketing department in any business is to monitor the company's performance, assess trends, and highlight any unusual results or anomalies. In a retail environment, they would keep track of the sales performance over time and identify any unusual rise or fall in results. This vital information allows the business to quickly rectify errors or capitalize on positive results.

In this exercise you will explore how you can use Microsoft Power BI features to quickly identify this type of crucial data. You will be asked to identify the two big sales spikes in the dataset and analyze them. You will apply the clustering technique to separate the data points into similar categories and then use the explain the increase tool of the **Analyze** feature to generate and add visualizations in your report and identify the driving factors behind this increase.

Case study

The marketing department at Adventure Works has identified an unexpected surge in bike sales on two distinct days in the previous month. Once they notified the senior management team of this sharp increase, the information caught the eye of the CEO. They have informed the marketing department that they are keen to identify the contributing factors behind this growth so that the company can capitalize on it and propel the business forward.

The manager of the marketing department has asked the analytics department to determine the reason for the surge in bike sales. It is important that the contributing factors are identified quickly as they may be time-specific and if so, the company would need to move quickly to take advantage of them and generate more business. After a brief discussion with your manager Jamie, you both feel that Microsoft Power BI's **Analyze** feature would be the most effective tool to rapidly generate visualizations that could uncover the driving forces behind the sales surge.

Instructions

Step 1: Download the data

- Download the Power BI report file titled *Adventure Works Sales Report.pbix* and open it in Power BI Desktop.

Step 2: Identify the two big sales spikes

To begin your analysis, you first examine the dataset you will be working with and create a proper visualization to identify the big sales days.

1. In **Data** view in Power BI, select the **Sales** table so you can familiarize yourself with the dataset.
2. Locate the **Order Date** column, **Order Total column**, as well as an attribute column that could support clustering. The **Product Name** column has high cardinality and is suitable for clustering.

3. Switch back to **Report** view and create a chart to depict a time series analysis, with the **Date** column in the **X-axis** and the **Amount** column in the **Y-axis**. Switch to the **Report** view and create a **Line chart** with **Order Date** on the **X-axis** and **Order Total** on the **Y-axis.** Opens in a new tab
4. With the help of the line chart, identify the two separate days where there was a spike in sales.

Step 3: Use the clustering technique to assist the Analyze feature

Before using the **Analyze** feature, it would be helpful to create a new product group using clustering techniques on the dataset.

1. Add a scatter chart with **Product Name** in the **Values** field, **Order Total** on the **X-axis**, and **Product Price** on the **Y-axis**.
2. Select three dots in the top right corner of this scatter chart and select **Automatically find clusters**.
3. Assign names and descriptions to your clusters and then enter **3** as the number of clusters.

Step 4: Use the Analyze feature to generate automated visualizations

After creating the clusters, it is time to use the Analyze feature on the days showing a sales surge to identify trends and patterns.

1. Identify the day with the most sales and use the **Explain the increase** tool of the analyze feature. Make a note of the percentile increase in the **Sum of Order Total**.
2. Note the five specific categories that seem to have had the most influence on the sales spike for that day.
3. Now pinpoint the day with the second highest number of sales and use the **Explain the increase** tool of the analyze feature on that day as well.
4. Using the analyze feature, identify which positive elements appear on this day as well. There are three shared contributory factors between the two days.

Step 5: Act on the insights by using the Analyze feature

The three fields mainly contributing to the sales increase have now been identified. You can now create visualizations and add them to the report.

1. Adjust the size of the two visualizations on the screen to an appropriate size for viewing.
2. It’s now time to add more insightful visualizations to the report. Switch back to **Data** view to identify the fields that will be added to visualizations.
3. Determine the number of distinct values in each field to aid you in the creation of the visualizations.
4. There is a field with two distinct values, a field with three distinct values, and a field with seven distinct values. Return to **Report** view and create two visualizations that properly depict a two or three value column and add them to your report.

Step 6: Act on the insights by Analyze feature

There is still one category to add to the report. You decide to use the **Analyze** feature again to review the impact of the result and add the last visualization to your report.

1. On the line chart, navigate back to the pop-up window produced by the **Explain the increase** choice in the **Analyze** feature for the 7th of March.
2. Scrolling through the generated visualizations, determine if any of them lack insights, and provide feedback to Power BI that it wasn’t helpful. This will improve the functioning of the **Analyze** feature in your future reports.

3. Locate the three most important factors for the increase and provide positive feedback to Power BI.
4. In the specific category that hasn't been visualized yet in the report, add the automatically generated visual to the report.
5. Now that all five visualizations are in the report, resize and customize them as per your preference before saving the report.

Conclusion

In this exercise, you used the **Analyze** feature to swiftly produce insightful results on your report. Walking through a practical example and following exactly the thought process of a data analyst, you were able to identify that the surge in bike sales was due to a high-selling performance of **Road Bikes** with a **Medium** product size. Working on this task you gained the experience of working with data and the power of the **Analyze** feature and the speed with which it generates results.